$$|1\rangle$$

$$|2\rangle$$

$$|3\rangle$$

$$|4\rangle$$

$$|5\rangle$$

$$|6\rangle$$

denoting the six faces of the die. Just like Alice's coin, Bob's die is quantum mechanical, and the six states can be superposed in a similar way.

6.1.2 Representing the Combined System

Now imagine that Bob's and Alice's systems both exist, and form a single composite system. The first question is: How could we construct the state-space—call it S_{AB}—for the combined system? The answer is to form the *tensor product* of S_A and S_B. The notation for this operation is

$$S_{AB} = S_A \otimes S_B.$$

To define S_{AB}, it is enough to specify its basis vectors. The basis vectors are exactly what you might expect. The top half of Fig. 6.1 shows a table whose columns correspond to Bob's six basis vectors and whose rows correspond to Alice's two basis vectors. Each box in the table denotes a basis vector for the S_{AB} system. For example, the box labeled $H4$ represents a state in S_{AB} in which the coin shows Heads and the die shows the number 4. In the combined system, there are twelve basis vectors altogether.